

E1987076_1

The equation $x^4 + x^2 + xy + y^3 + 2 = 0$ has no integer solutions

Size $h = 34$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + x^2 + xy + y^3 + 2 = 0 \tag{1}$$

has no integer solutions.

The proof uses two polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), with $(d_1, d_2) = (-2, -210)$, and the fact that an odd prime p with $(-d_k/p) = -1$ divides $U_k^2 + d_k V_k^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Jacobi symbol and the law of quadratic reciprocity.

1 Jacobi symbols and the norm forms $a^2 + db^2$

Throughout this section, d is a squarefree integer with $d \neq 0, -1$; we apply the results with $d \in \{-210, -2\}$. For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. For an odd positive integer n , the Jacobi symbol is $(a/n) = \prod_p (a/p)^{v_p(n)}$, the product over the prime factors p of n . It is multiplicative in a and in n , it depends only on a modulo n , and it is non-zero exactly when a and n are coprime. We use the law of quadratic reciprocity for Jacobi symbols together with its supplements [1, Chapter 5, §2]: for odd positive integers m and n ,

$$\begin{aligned} \left(\frac{-1}{n}\right) &= (-1)^{(n-1)/2}, & \left(\frac{2}{n}\right) &= (-1)^{(n^2-1)/8}, \\ \left(\frac{m}{n}\right) \left(\frac{n}{m}\right) &= (-1)^{(m-1)(n-1)/4} \quad \text{if } \gcd(m, n) = 1. \end{aligned}$$

Lemma 2. *Let p be an odd prime with $(-d/p) = -1$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + db^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$. Then $(ac)^2 \equiv -d(bc)^2 \equiv -d \pmod{p}$, so $-d$ is a square modulo p . As $(-d/p) \neq 0$, it is a non-zero square, so $(-d/p) = 1$, a contradiction. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let $a, b \in \mathbb{Z}$ be not both zero. Then $a^2 + db^2 \neq 0$, and $v_p(a^2 + db^2)$ is even for every odd prime p with $(-d/p) = -1$.*

Proof. If $d > 0$, then clearly $a^2 + db^2 > 0$. If $d < 0$, then $-d > 1$ is squarefree, hence not a square, so $a^2 + db^2 \neq 0$. Let p be an odd prime with $(-d/p) = -1$, let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + db'^2$ by Lemma 2. Since $a^2 + db^2 = p^{2k}(a'^2 + db'^2)$, we get $v_p(a^2 + db^2) = 2k$. $\square$

Lemma 4. *Let n and n' be odd positive integers with $n' \equiv n \pmod{4|d|}$, or, if $d < 0$, with $n' \equiv -n \pmod{4|d|}$. Then $(-d/n') = (-d/n)$.*

Proof. Since d is squarefree, we can write $-d = \varepsilon 2^e q$ with $\varepsilon \in \{1, -1\}$, $e \in \{0, 1\}$ and q odd and positive. By multiplicativity and the reciprocity laws,

$$\left(\frac{-d}{n}\right) = \left(\frac{\varepsilon}{n}\right) \left(\frac{2}{n}\right)^e (-1)^{(n-1)(q-1)/4} \left(\frac{n}{q}\right).$$

Here (ε/n) and $(-1)^{(n-1)(q-1)/4}$ depend only on n modulo 4, $(2/n)$ depends only on n modulo 8, and (n/q) depends only on n modulo q . Since $4|d|$ is divisible by $4q$, and by $8q$ if $e = 1$, the claim for $n' \equiv n$ follows. If $d < 0$ and $n' \equiv -n \pmod{4|d|}$, then, as $\varepsilon = 1$, the factor (ε/n) does not change, $(2/n)$ does not change because $n'^2 \equiv n^2 \pmod{16}$, and each of $(-1)^{(n-1)(q-1)/4}$ and (n/q) is multiplied by $(-1)^{(q-1)/2}$, because $(n' - 1)/2 \equiv (n - 1)/2 + 1 \pmod{2}$ and $(n'/q) = (-1/q)(n/q) = (-1)^{(q-1)/2}(n/q)$. $\square$

Lemma 5. *Let n be a non-zero integer, and let s, u and m be positive integers such that every prime factor of s divides $2|d|$, both $2s$ and $4|d|s$ divide m , u is odd, $(-d/u) = -1$, and $n \equiv su \pmod{m}$. If $d > 0$, suppose also that $n > 0$. Then there is an odd prime p with $(-d/p) = -1$ such that $v_p(n)$ is odd.*

Proof. Since $s \mid m$, we have $s \mid n$, and $n' = n/s$ satisfies $n' \equiv u \pmod{m/s}$. Since m/s is even and divisible by $4|d|$, n' is odd and $n' \equiv u \pmod{4|d|}$. If $d > 0$, then $n' > 0$, and if $d < 0$, then $|n'| \equiv \pm u \pmod{4|d|}$; in both cases Lemma 4 gives $(-d/|n'|) = (-d/u) = -1$. By multiplicativity, the product of $(-d/p)^{v_p(n')}$ over the prime factors p of $|n'|$ is -1 , so some prime factor p of n' satisfies $(-d/p) = -1$ and $v_p(n')$ is odd. Then p is odd, and $p \nmid 2d$, so $p \nmid s$ and $v_p(n) = v_p(n')$ is odd. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + x^2 + xy + y^3 + 2$, so that (1) is the equation $F(x, y) = 0$. For the first identity, $d_1 = -2$, and we define

$$\begin{aligned} H_1 &= -4x - 2y + 1, \\ T_1 &= 32x^2 - 16xy - 8x + 8y^2 + 4y + 2, \\ U_1 &= 4x^2 - 16x + 14, \\ V_1 &= 12x - 9, \\ G_1 &= -16, \\ A_1 &= 12, \\ B_1 &= -4x + 13, \end{aligned}$$

for the second identity, $d_2 = -210$, and we define

$$\begin{aligned} H_2 &= 20x - 10y - 19, \\ T_2 &= 4000x^2 + 2000xy - 6600x + 1000y^2 - 1900y + 3610, \\ U_2 &= 100x^2 + 400x - 110, \\ V_2 &= 40x - 17, \\ G_2 &= -10000, \\ A_2 &= 16, \\ B_2 &= -40x - 177. \end{aligned}$$

Expanding both sides, one checks the identities

$$H_1 T_1 - U_1^2 + 2V_1^2 = G_1 F, \tag{2}$$

$$H_2T_2 - U_2^2 + 210V_2^2 = G_2F, \quad (3)$$

$$A_1U_1 + B_1V_1 = 3 \cdot 17, \quad (4)$$

$$A_2U_2 + B_2V_2 = 1249. \quad (5)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) and (3) give

$$H_kT_k = U_k^2 + d_kV_k^2 \quad (k = 1, 2), \quad (6)$$

where the polynomials now denote their values at (x, y) .

Step 1: congruences. Checking the 2520^2 residue pairs modulo 2520, we find that (1) has exactly 768 solutions modulo 2520, and that at each of them $H_1 \bmod 2520$ lies in the set $\mathcal{R}_1$ listed in Table 1 or $H_2 \bmod 2520$ lies in the set $\mathcal{R}_2$ listed in Table 1. For $k = 1$, each $r \in \mathcal{R}_1$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(2/u) = -1$; here for $s = 1$, $2s = 2$ and $8s = 8$ divide 2520. For $k = 2$, each $r \in \mathcal{R}_2$ is listed as $r = su$ with s and u positive and u odd, or as r if $s = 1$, and in each case $(210/u) = -1$; here for $s = 1$, $2s = 2$ and $840s = 840$ divide 2520, and for $s = 3$, every prime factor of s divides 420, and $2s = 6$ and $840s = 2520$ divide 2520.

Step 2: signs. *The value of H_1 .* By (4), U_1 and V_1 are not both zero, so $U_1^2 - 2V_1^2 \neq 0$ by Lemma 3 with $d = -2$, and (6) gives $H_1 \neq 0$.

The value of H_2 . By (5), U_2 and V_2 are not both zero, so $U_2^2 - 210V_2^2 \neq 0$ by Lemma 3 with $d = -210$, and (6) gives $H_2 \neq 0$.

Step 3: primes p with $(-d_k/p) = -1$. Let $k \in \{1, 2\}$, and let p be an odd prime with $(-d_k/p) = -1$, where $(d_1, d_2) = (-2, -210)$. We show that $v_p(H_k)$ is even. By (4)–(5), U_k and V_k are not both zero, so (6) and Lemma 3 show that $v_p(H_kT_k) = v_p(U_k^2 + d_kV_k^2)$ is even. Suppose that p divides both H_k and T_k . Then $p \mid U_k^2 + d_kV_k^2$ by (6), so $p \mid U_k$ and $p \mid V_k$ by Lemma 2, and $p \mid R_k$, where R_k is the right-hand side of the identity for $A_kU_k + B_kV_k$. For $k = 1$, we have $R_1 = 3 \cdot 17$. As p is odd and $p \neq 17$ because $(2/17) = 1$, we get $p = 3$. Checking the 3^2 residue pairs modulo 3, we find that F , H_1 and T_1 have no common zero modulo 3, a contradiction. For $k = 2$, we have $R_2 = 1249$. As p is odd, we get $p = 1249$. Checking the 1249^2 residue pairs modulo 1249, we find that F , H_2 and T_2 have no common zero modulo 1249, a contradiction. Hence p does not divide both H_k and T_k . If $p \nmid T_k$, then $v_p(H_k) = v_p(H_kT_k)$ is even, and if $p \nmid H_k$, then $v_p(H_k) = 0$.

Step 4: contradiction. By Step 1, there is k such that $H_k \equiv su \pmod{2520}$ with s and u as in Lemma 5 for $d = d_k$. By Step 2, $H_k \neq 0$, and $H_k > 0$ if $d_k > 0$. Hence Lemma 5 applies to $n = H_k$ and $d = d_k$, so there is an odd prime p with $(-d_k/p) = -1$ such that $v_p(H_k)$ is odd. This contradicts Step 3, and the theorem is proved. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

Table 1: The sets $\mathcal{R}_k$ of residues modulo 2520 in Step 1.

$\mathcal{R}_1$, the residues of H_1					
37	461	877	1301	1717	2141
77	517	917	1357	1757	2197
91	541	931	1381	1771	2221
107	581	947	1421	1787	2261
181	587	1021	1427	1861	2267
251	611	1091	1451	1931	2291
301	637	1141	1477	1981	2317
307	667	1147	1507	1987	2347
317	707	1157	1547	1997	2387
331	797	1171	1637	2011	2477
371	811	1211	1651	2051	2491
427	821	1267	1661	2107	2501
$\mathcal{R}_2$, the residues of H_2					
$51 = 3 \cdot 17$	331	1051	1891		
211	$771 = 3 \cdot 257$	1171	2011		